

Multiple Choice

1. **Answer:** E

Options: A) The heart synthesizes its own nutrients internally without reliance on external sources.; B) The heart relies on neural control for its metabolic needs.; C) The heart extracts nutrients from the atria and ventricles.; D) The heart's metabolic needs are met by the lymphatic system.; E) The heart depends on its own blood supply from the coronary vessels.

Note: The heart relies on the coronary arteries, which branch off from the aorta, to deliver oxygenated blood and essential nutrients directly to its tissues, as its chambers do not extract these substances from the blood passing through them.

2. **Answer:** B

Options: A) a large population; B) genetic drift; C) random mating; D) absence of selection

Note: Genetic drift is not a condition assumed by the Hardy-Weinberg equilibrium; rather, the equilibrium assumes a large population size where genetic drift has minimal effect, allowing allele frequencies to remain constant over generations.

3. **Answer:** A

Options: A) The drop in blood pressure causes blood clotting; B) The drop in blood pressure causes an increase in plasma protein concentration; C) The drop in blood pressure stimulates the excretion of fluids by the kidneys; D) The drop in blood pressure causes dehydration; E) The drop in blood pressure triggers the release of antidiuretic hormone

Note: The drop in blood pressure reduces the hydrostatic pressure within capillaries, promoting fluid retention through increased osmotic pressure and facilitating blood clotting as a physiological response to maintain hemostasis and fluid balance.

4. **Answer:** D

Options: A) Coevolution; B) Punctuated equilibrium; C) Directional selection; D) Divergent evolution; E) Convergent evolution

Note: Divergent evolution occurs when two related species evolve different traits or behaviors, often due to adapting to different environments or ecological niches, leading to a reduction in their similarities.

5. **Answer:** A

Options: A) positive chemotropism; B) positive phototropism; C) negative phototropism; D) negative chemotropism; E) positive hydrotropism

Note: The correct answer is positive chemotropism because plant growth towards or away from a stimulus, such as gravity, is influenced by chemical signals that direct the growth response, indicating a positive response to the gravitational force.

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Both robins and dogs are warm-blooded, or endothermic, animals that have the physiological ability to regulate and maintain a stable internal body temperature despite variations in their external environment.

7. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the synthesis of an RNA/DNA hybrid requires reverse transcriptase to accurately synthesize complementary DNA from an RNA template, as RNase alone cannot catalyze this reaction.

8. **Answer:** True

Options: A) True; B) False

Note: Coevolution is true because it describes the process where two or more species exert selective pressures on each other, leading to reciprocal adaptations, particularly evident in interactions like predator-prey dynamics.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the wings of butterflies and birds are not homologous structures; they evolved independently from different ancestral origins, making them examples of convergent evolution rather than shared ancestry.

10. **Answer:** False

Options: A) True; B) False

Note: Centrosomes are primarily involved in organizing microtubules and facilitating chromosome separation during cell division, while the regulation of cell division checkpoints is primarily controlled by proteins and signaling pathways, such as cyclins and cyclin-dependent kinases.

Long Form Response

11. **Answer:** Ontogeny in humans begins with the zygote, a single fertilized cell that undergoes a series of rapid divisions to form a blastocyst, which then implants into the uterine wall. This marks the embryonic stage, where the foundational structures and organs begin to develop through processes such as gastrulation and organogenesis. As the embryo transitions into the fetal stage, significant growth occurs, with further differentiation of tissues and systems, ultimately leading to a mature fetus. Each stage of development is crucial, as it lays the groundwork for physical and functional capabilities, influencing overall human health and growth. Understanding these stages highlights the complexity of human development and the importance of a supportive environment for optimal outcomes.

Note: correct because it accurately outlines the key stages of human ontogeny, emphasizing the critical processes of cell division, implantation, gastrulation, and organogenesis, as well as the transition to the fetal stage, all of which are essential for understanding human growth and health.

12. **Answer:** The fate of a vesicle formed through phagocytosis is that it remains intact and floats freely in the cytoplasm. This vesicle, often referred to as a phagosome, can interact with lysosomes to form a phagolysosome where the engulfed material is degraded. The intact vesicle allows for the potential release of signaling molecules and nutrients into the cytoplasm, influencing cellular processes such as immune responses and metabolic activities. Additionally, the presence of the vesicle can trigger various cellular signaling pathways, impacting gene expression and cellular function. Overall, the integrity of the vesicle plays a crucial role in the effective processing of ingested materials and the maintenance of cellular homeostasis.

Note: correct because it accurately describes the transformation of a phagosome into a phagolysosome for degradation, highlights the significance of the vesicle in nutrient and signaling molecule release, and connects these processes to broader cellular functions such as immune responses and metabolism.